

Администрирование локальных сетей

Лабораторная работа №2

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Цели и задачи работы

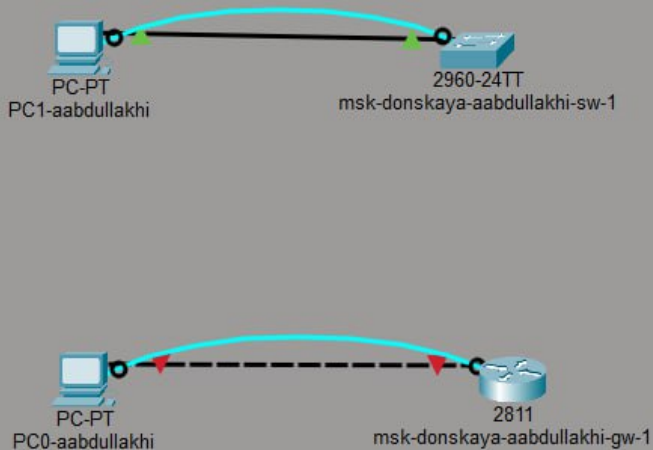
Цель лабораторной работы

Целью данной лабораторной работы является приобретение практических навыков по первичной настройке сетевого оборудования Cisco, изучение базовых команд операционной системы Cisco IOS, а также освоение принципов организации удалённого доступа к маршрутизатору и коммутатору.

Задачи лабораторной работы

- Назначение IP-адресов
- Конфигурация интерфейсов
- Создание VLAN
- Настройка паролей доступа
- Организация удалённого управления по SSH
- Проверка соединений с использованием команды ping

Выполнение лабораторной работы



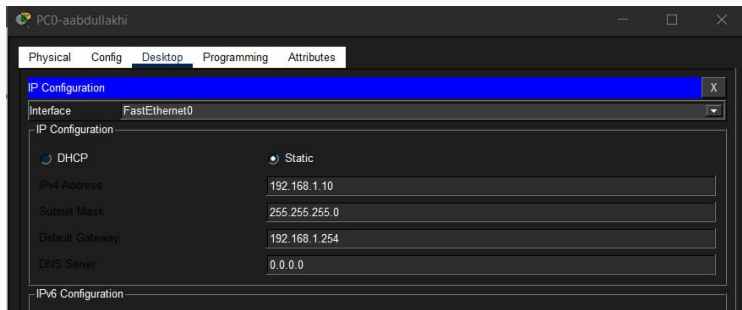


Рис. 2: Настройка IP-адреса на PC0

Настройка маршрутизатора

```
msk-donskaya-aabdullakhi-gw-1
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname msk-donskaya-aabdullakhi-gw-1
msk-donskaya-aabdullakhi-gw-1(config)#interface f0/0
msk-donskaya-aabdullakhi-gw-1(config-if)#no shutdown

msk-donskaya-aabdullakhi-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

msk-donskaya-aabdullakhi-gw-1(config-if)#ip address 192.168.1.254 255.255.255.0
msk-donskaya-aabdullakhi-gw-1(config-if)#line vty 0 4
msk-donskaya-aabdullakhi-gw-1(config-line)#password cisco
msk-donskaya-aabdullakhi-gw-1(config-line)#login
msk-donskaya-aabdullakhi-gw-1(config-line)#line console 0
msk-donskaya-aabdullakhi-gw-1(config-line)#password cisco
msk-donskaya-aabdullakhi-gw-1(config-line)#login
msk-donskaya-aabdullakhi-gw-1(config-line)#enable secret cisco
msk-donskaya-aabdullakhi-gw-1(config)#service password-encryption
msk-donskaya-aabdullakhi-gw-1(config)#username admin privilege 1 secret cisco
msk-donskaya-aabdullakhi-gw-1(config)#ip domain-name donskeya.rudn.edu
msk-donskaya-aabdullakhi-gw-1(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-aabdullakhi-gw-1.donskeya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 512
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-aabdullakhi-gw-1(config)#line vty 0 4
*Mar 1 0:21:24.511: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:21:24.511: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-aabdullakhi-gw-1(config-line)#transport input ssh
msk-donskaya-aabdullakhi-gw-1(config-line)#exit
msk-donskaya-aabdullakhi-gw-1(config)#exit
msk-donskaya-aabdullakhi-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-aabdullakhi-gw-1#write memory
Building configuration...
[OK]
msk-donskaya-aabdullakhi-gw-1#
```

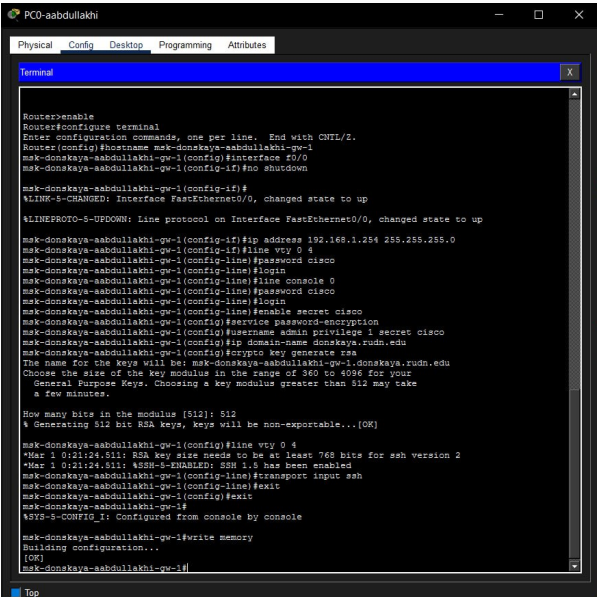
Активация Windows

Чтобы активировать Windows, перейдите в раз

Copy Paste

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Настройка маршрутизатора через консоль



```
PC0-aabdullakhi
Physical Config Desktop Programming Attributes

Terminal

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTRL/Z.
Router(config)#hostname msk-donskaya-aabdullakhi-gw-1
msk-donskaya-aabdullakhi-gw-1(config)#interface f0/0
msk-donskaya-aabdullakhi-gw-1(config-if)#no shutdown

msk-donskaya-aabdullakhi-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

msk-donskaya-aabdullakhi-gw-1(config-if)#ip address 192.168.1.254 255.255.255.0
msk-donskaya-aabdullakhi-gw-1(config-if)#line vty 0 4
msk-donskaya-aabdullakhi-gw-1(config-line)#password cisco
msk-donskaya-aabdullakhi-gw-1(config-line)#login
msk-donskaya-aabdullakhi-gw-1(config-line)#line console 0
msk-donskaya-aabdullakhi-gw-1(config-line)#password cisco
msk-donskaya-aabdullakhi-gw-1(config-line)#login
msk-donskaya-aabdullakhi-gw-1(config-line)#enable secret cisco
msk-donskaya-aabdullakhi-gw-1(config)#service password-encryption
msk-donskaya-aabdullakhi-gw-1(config)#username admin privilege 1 secret cisco
msk-donskaya-aabdullakhi-gw-1(config)#ip domain-name donskeya.rudn.edu
msk-donskaya-aabdullakhi-gw-1(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-aabdullakhi-gw-1.donskeya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 512
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-aabdullakhi-gw-1(config)#line vty 0 4
*Mar 1 0:21:24.511: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:21:24.511: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-aabdullakhi-gw-1(config-line)#transport input ssh
msk-donskaya-aabdullakhi-gw-1(config-line)#exit
msk-donskaya-aabdullakhi-gw-1(config)#exit
msk-donskaya-aabdullakhi-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-aabdullakhi-gw-1#write memory
Building configuration...
[OK]
msk-donskaya-aabdullakhi-gw-1#
```

Проверка соединения с маршрутизатором

```
PC0-aabdullakhi
Physical Config Desktop Programming Attributes
Command Prompt X
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.254

Pinging 192.168.1.254 with 32 bytes of data:

Reply from 192.168.1.254: bytes=32 time<1ms TTL=255
Reply from 192.168.1.254: bytes=32 time<1ms TTL=255
Reply from 192.168.1.254: bytes=32 time<1ms TTL=255
Reply from 192.168.1.254: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.254:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>telnet 192.168.1.254
Trying 192.168.1.254 ...Open

[Connection to 192.168.1.254 closed by foreign host]
C:\>ssh -l admin 192.168.1.254
Invalid Command.

C:\>ssh 192.168.1.254
Invalid Command.

C:\>ssh -l admin 192.168.1.254
Password:
Password:
Password:

[Connection to 192.168.1.254 closed by foreign host]
C:\>ssh -l admin 192.168.1.254
Password:

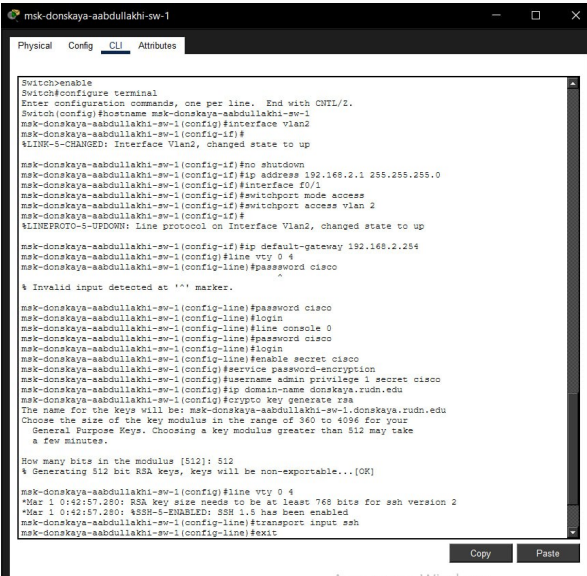
msk-donskaya-aabdullakhi-gw-1>
msk-donskaya-aabdullakhi-gw-1>
msk-donskaya-aabdullakhi-gw-1>exit

[Connection to 192.168.1.254 closed by foreign host]
C:\>
```

Активация Windows
Чтобы активировать Windows, перейдите в раздел
"Параметры".

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Настройка коммутатора Cisco 2960



```
msk-donskaya-aabdullakhi-sw-1
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTRL/Z.
Switch(config)#hostname msk-donskaya-aabdullakhi-sw-1
msk-donskaya-aabdullakhi-sw-1(config)#interface vlan2
msk-donskaya-aabdullakhi-sw-1(config-if)#
%LINK-S-CHANGED: Interface Vlan2, changed state to up

msk-donskaya-aabdullakhi-sw-1(config-if)#no shutdown
msk-donskaya-aabdullakhi-sw-1(config-if)#ip address 192.168.2.1 255.255.255.0
msk-donskaya-aabdullakhi-sw-1(config-if)#interface f0/1
msk-donskaya-aabdullakhi-sw-1(config-if)#switchport mode access
msk-donskaya-aabdullakhi-sw-1(config-if)#switchport access vlan 2
msk-donskaya-aabdullakhi-sw-1(config-if)#
%LINEPROTO-S-UPDOWN: Line protocol on Interface Vlan2, changed state to up

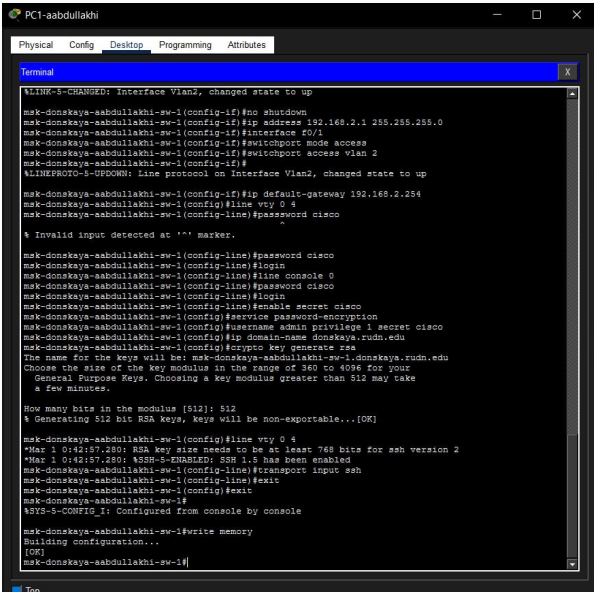
msk-donskaya-aabdullakhi-sw-1(config-if)#ip default-gateway 192.168.2.254
msk-donskaya-aabdullakhi-sw-1(config)#line vty 0 4
msk-donskaya-aabdullakhi-sw-1(config-line)#password cisco
^
% Invalid input detected at '^' marker.

msk-donskaya-aabdullakhi-sw-1(config-line)#password cisco
msk-donskaya-aabdullakhi-sw-1(config-line)#login
msk-donskaya-aabdullakhi-sw-1(config-line)#line console 0
msk-donskaya-aabdullakhi-sw-1(config-line)#password cisco
msk-donskaya-aabdullakhi-sw-1(config-line)#login
msk-donskaya-aabdullakhi-sw-1(config-line)#enable secret cisco
msk-donskaya-aabdullakhi-sw-1(config)#service password-encryption
msk-donskaya-aabdullakhi-sw-1(config)#username admin privilege 1 secret cisco
msk-donskaya-aabdullakhi-sw-1(config)#ip domain-name donskeya.rudn.edu
msk-donskaya-aabdullakhi-sw-1(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-aabdullakhi-sw-1.donskeya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 512
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-aabdullakhi-sw-1(config)#line vty 0 4
*Mar 1 0:42:57.280: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:42:57.280: %SSH-S-ENABLED: SSH 1.5 has been enabled
msk-donskaya-aabdullakhi-sw-1(config-line)#transport input ssh
msk-donskaya-aabdullakhi-sw-1(config-line)#exit
```

Настройка коммутатора через консоль



```
PC1-aabdullakhi
Physical Config Desktop Programming Attributes

Terminal

%LINK-5-CHANGED: Interface Vlan2, changed state to up

msk-donskaya-aabdullakhi-sw-1(config-if)#no shutdown
msk-donskaya-aabdullakhi-sw-1(config-if)#ip address 192.168.2.1 255.255.255.0
msk-donskaya-aabdullakhi-sw-1(config-if)#interface f0/1
msk-donskaya-aabdullakhi-sw-1(config-if)#switchport mode access
msk-donskaya-aabdullakhi-sw-1(config-if)#switchport access vlan 2
msk-donskaya-aabdullakhi-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to up

msk-donskaya-aabdullakhi-sw-1(config-if)#ip default-gateway 192.168.2.254
msk-donskaya-aabdullakhi-sw-1(config)#line vty 0 4
msk-donskaya-aabdullakhi-sw-1(config-line)#password cisco
^
% Invalid input detected at '^' marker.

msk-donskaya-aabdullakhi-sw-1(config-line)#password cisco
msk-donskaya-aabdullakhi-sw-1(config-line)#login
msk-donskaya-aabdullakhi-sw-1(config-line)#line console 0
msk-donskaya-aabdullakhi-sw-1(config-line)#password cisco
msk-donskaya-aabdullakhi-sw-1(config-line)#login
msk-donskaya-aabdullakhi-sw-1(config-line)#enable secret cisco
msk-donskaya-aabdullakhi-sw-1(config)#service password-encryption
msk-donskaya-aabdullakhi-sw-1(config)#username admin privilege 1 secret cisco
msk-donskaya-aabdullakhi-sw-1(config)#ip domain-name donskeya.rudn.edu
msk-donskaya-aabdullakhi-sw-1(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-aabdullakhi-sw-1.donskeya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 512
% Generating 512 bit RSA Keys, Keys will be non-exportable...[OK]

msk-donskaya-aabdullakhi-sw-1(config)#line vty 0 4
*Mar 1 0:42:57.280: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:42:57.280: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-aabdullakhi-sw-1(config-line)#transport input ssh
msk-donskaya-aabdullakhi-sw-1(config-line)#exit
msk-donskaya-aabdullakhi-sw-1(config)#exit
msk-donskaya-aabdullakhi-sw-1#
%SYS-5-CONFIG I: Configured from console by console

msk-donskaya-aabdullakhi-sw-1#write memory
Building configuration...
[OK]
msk-donskaya-aabdullakhi-sw-1#
```

Настройка PC1

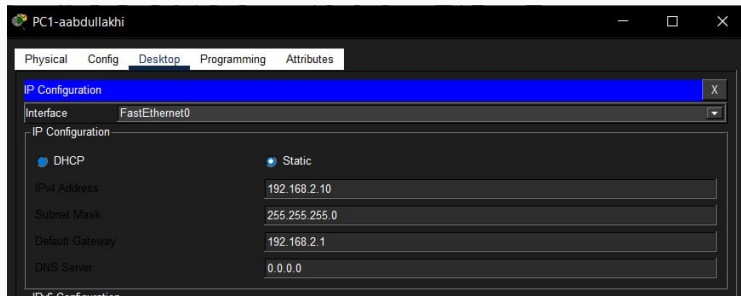
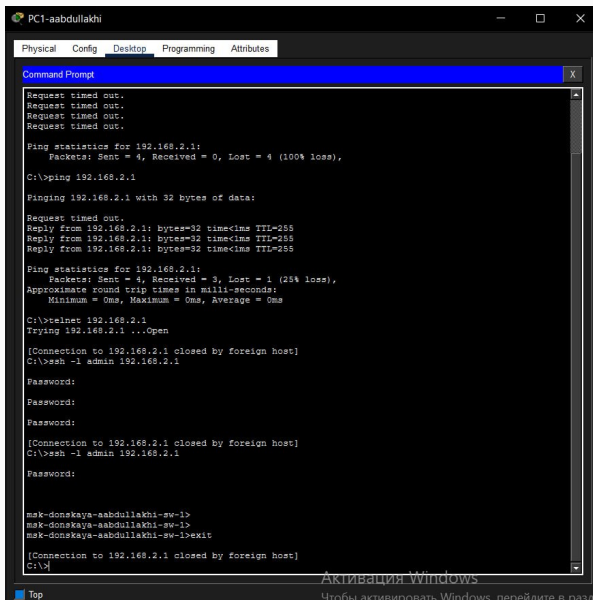


Рис. 8: Настройка IP-адреса на PC1

Проверка соединения с коммутатором



The screenshot shows a Windows Command Prompt window titled "PC1-aabdullakhi". The window has tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes", with "Config" selected. The Command Prompt displays the following text:

```
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>telnet 192.168.2.1
Trying 192.168.2.1 ...Open

[Connection to 192.168.2.1 closed by foreign host]
C:\>ssh -l admin 192.168.2.1

Password:
Password:
Password:

[Connection to 192.168.2.1 closed by foreign host]
C:\>ssh -l admin 192.168.2.1

Password:

msk-donskaya-aabdullakhi-sw-1>
msk-donskaya-aabdullakhi-sw-1>
msk-donskaya-aabdullakhi-sw-1>exit

[Connection to 192.168.2.1 closed by foreign host]
C:\>
```

At the bottom of the window, there is a watermark that reads "Активация Windows" (Windows Activation) and a small "Top" button.

Выводы лабораторной работы

В ходе лабораторной работы Абдуллахи Абдул Вахид выполнил сборку и настройку простой сети в Cisco Packet Tracer, состоящей из маршрутизатора, коммутатора и двух персональных компьютеров. Были успешно освоены базовые навыки:

- назначение IP-адресов
- конфигурация интерфейсов
- создание VLAN
- настройка паролей доступа
- организация удалённого управления по SSH

Проверка соединений с использованием команды ping показала корректную работу сети. Настроенные механизмы удалённого доступа функционируют стабильно, что подтверждает правильность выполненной конфигурации.